

Question Number	Answer	Additional guidance	Mark
8(a)	An explanation that includes two of the following points: • haploid so that when it is fertilised {a zygote is formed / it becomes diploid} (1) • contains lipid droplets as a source of energy (1) • {cortical granules / enzymes} to {harden zona pellucida / prevent polyspermy} (1)	ACCEPT large cell to store more {lipid/energy} ACCEPT releases chemicals to attract sperm ACCEPT sperm bind to glycoproteins on egg cell (surface)	(2)

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8(b)(i)	An answer that includes the following points: Similarities • lining up of {chromosomes/ chromatids} on equator (of cell) (1) • centromere bonds to spindle fibres (in metaphase) (1) Differences (max 2): • two metaphase stages in meiosis whereas mitosis has one (1) • {independent/random} assortment occurs (in metaphase I) in meiosis whereas it does not occur in mitosis (1) • meiosis involves homologous pairs of chromosomes (in metaphase I) and sister chromatids (in metaphase II) whereas mitosis involves sister chromatids (1)	Full marks can only be awarded if there is a similarity in the answer ACCEPT centre of cell / metaphase plate ACCEPT {chromosomes/chromatids} bond to spindle fibres ACCEPT independent assortment only occurs in metaphase I in meiosis ACCEPT meiosis involves pairs of chromosomes whereas mitosis involves pairs of {sister chromatids/ single chromosomes} ACCEPT meiosis involves pairs of chromosomes whereas mitosis does not	(3)

Question Number	Answer	Additional guidance	Mark
8(b)(ii)	An explanation that includes four of the following points: • morula cells are totipotent whereas blastocyst cells are pluripotent (1) • (specialisation / differentiation occurs because) gene(s) switched off (in blastocyst cells) (1) • due to {epigenetic modification / DNA methylation / histone modification} (1) • transcription of (active) genes to produce (active) mRNA / (active) mRNA translated at the ribosomes (1) • the protein(s) produced {determine / change} cell {structure / function} (1)	ACCEPT cells become pluripotent ACCEPT differential gene expression occurs ACCEPT transcription and translation of (active)gene to produce protein	(4)

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8(c)(i)	- correct D2 values (1) - correct calculation of $\sum D^2$ (1)	<table><tr><th>Age (weeks)</th><th>Rank</th><th>Mass of egg (g)</th><th>Rank</th><th>D</th><th>D²</th></tr><tr><td>36</td><td>1</td><td>2.27</td><td>8</td><td>-7</td><td>49</td></tr><tr><td>39</td><td>2</td><td>1.91</td><td>1</td><td>1</td><td>1</td></tr><tr><td>42</td><td>3</td><td>2.18</td><td>5.5</td><td>-2.5</td><td>6.25</td></tr><tr><td>45</td><td>4</td><td>2.28</td><td>9</td><td>-5</td><td>25</td></tr><tr><td>48</td><td>5</td><td>2.12</td><td>4</td><td>1</td><td>1</td></tr><tr><td>51</td><td>6</td><td>2.19</td><td>7</td><td>-1</td><td>1</td></tr><tr><td>54</td><td>7</td><td>2.18</td><td>5.5</td><td>1.5</td><td>2.25</td></tr><tr><td>57</td><td>8</td><td>2.09</td><td>3</td><td>5</td><td>25</td></tr><tr><td>60</td><td>9</td><td>2.03</td><td>2</td><td>7</td><td>49</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td>159.5</td></tr></table> $\sum D^2 = 159.5$ Correct answer with no working shown scores full marks 1 mark awarded for correct $\sum$ using incorrect D² values calculated in table	Age (weeks)	Rank	Mass of egg (g)	Rank	D	D ²	36	1	2.27	8	-7	49	39	2	1.91	1	1	1	42	3	2.18	5.5	-2.5	6.25	45	4	2.28	9	-5	25	48	5	2.12	4	1	1	51	6	2.19	7	-1	1	54	7	2.18	5.5	1.5	2.25	57	8	2.09	3	5	25	60	9	2.03	2	7	49						159.5	(2)
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8(c)(ii)	- correct values substituted into formula (1) - correct r_s value (1)	Example of calculation: $957 \div 720$ $= -0.329 / -0.33 / -0.3$ Allow ECF from answer calculated in (c)(i) Correct answer with no working shown scores full marks	(2)

Question Number	Answer	Additional guidance	Mark
8(c)(iii)	An explanation that includes the following points: • (the null hypothesis) should be accepted (1) • (because) 0.38 is lower than 0.683 (1)	mark points independently	(2)